

Simulation Modeling and Performance Evaluation

Lecture 14 **Input Data Modeling**

Muhammad Mahbub Alam, PhD
Professor
CSE Department, IUT

Introduction

- Every simulation requires one or more input data
 - e.g., interarrival times of customer, demand amount for an inventory, etc.
- Most of the cases, the input data are random
- To understand the system behavior, a huge amount of data is required
- Input data modeling
 - Makes use of probabilistic distribution
 - Either a theoretical distribution or an empirical one
- In contrast, input data can be collected from real systems.

Sources of randomness for common applications

TABLE 6.1

Sources of randomness for common simulation applications

Type of system	Sources of randomness
Manufacturing	Processing times, machine times to failure, machine repair times
Defense-related	Arrival times and payloads of missiles or airplanes, outcome of an engagement, miss distances for munitions
Communications	Interarrival times of messages, message types, message lengths
Transportation	Ship-loading times, interarrival times of customers to a subway

Example data - Processing Times

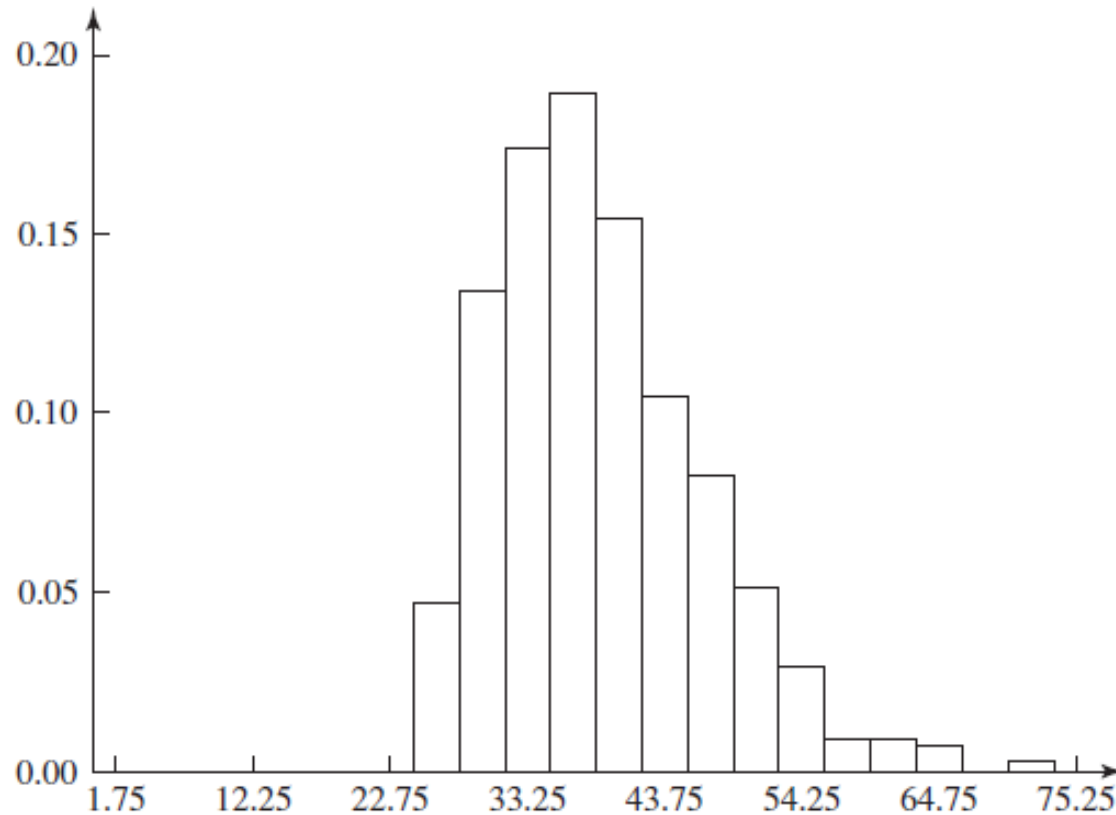


FIGURE 6.1

Histogram of 890 machine processing times for an automotive manufacturer.

Example Data – Interarrival Times

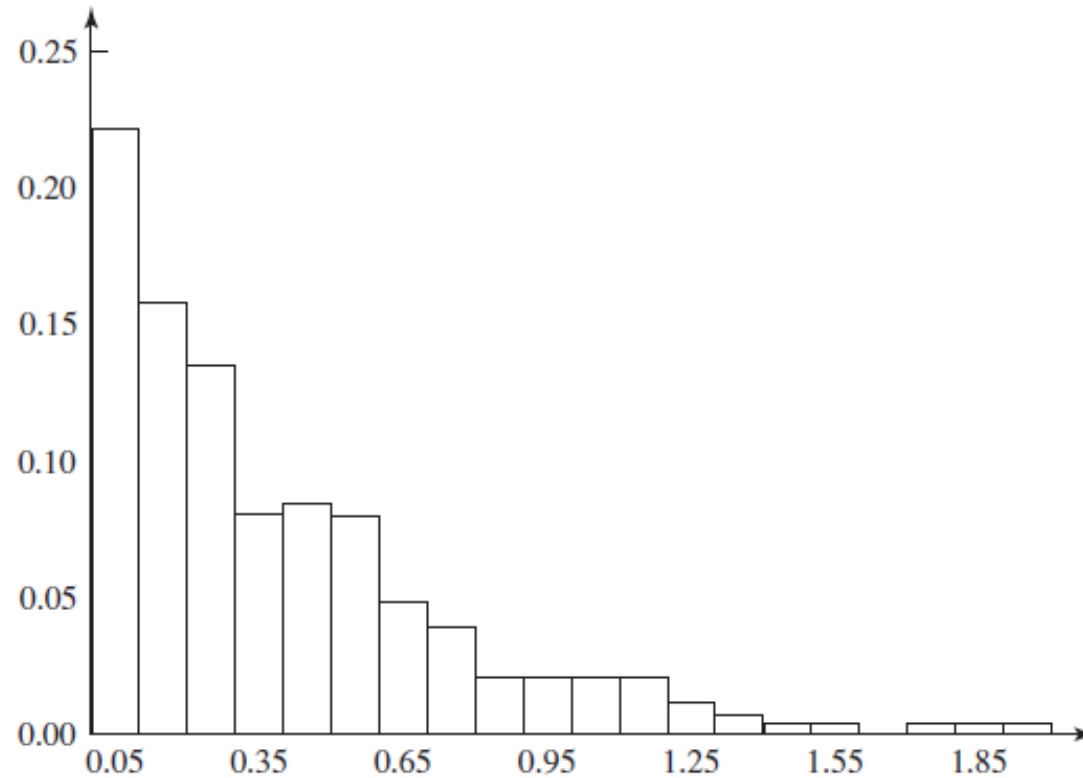


FIGURE 6.2
Histogram of 219 interarrival times to a drive-up bank.

Example Data – Ship-Loading Times

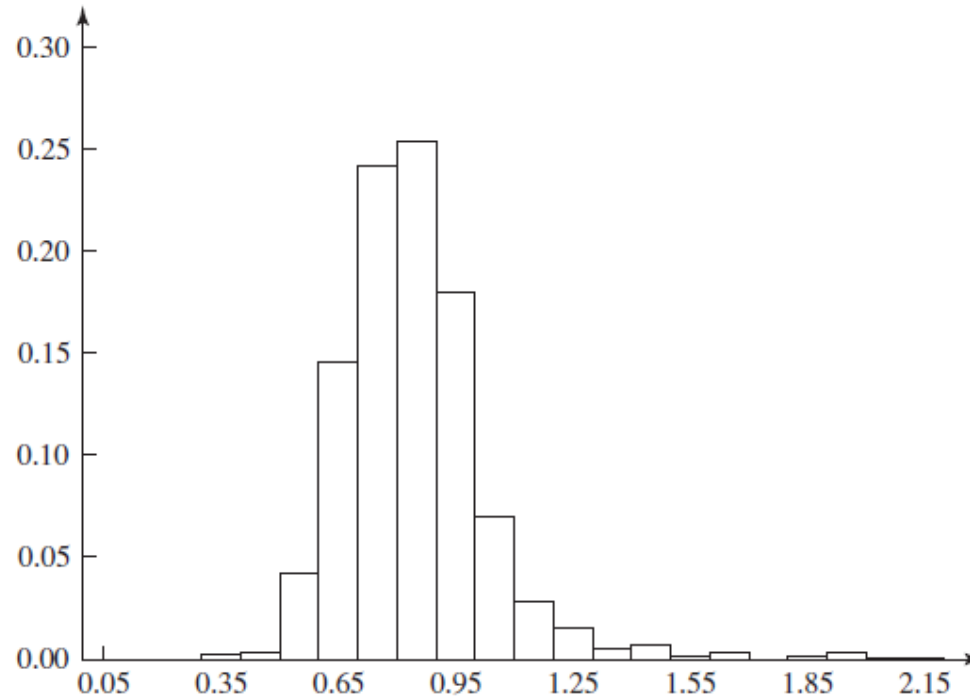


FIGURE 6.3
Histogram of 856 ship-loading times.

Example Data – Length of Tissue

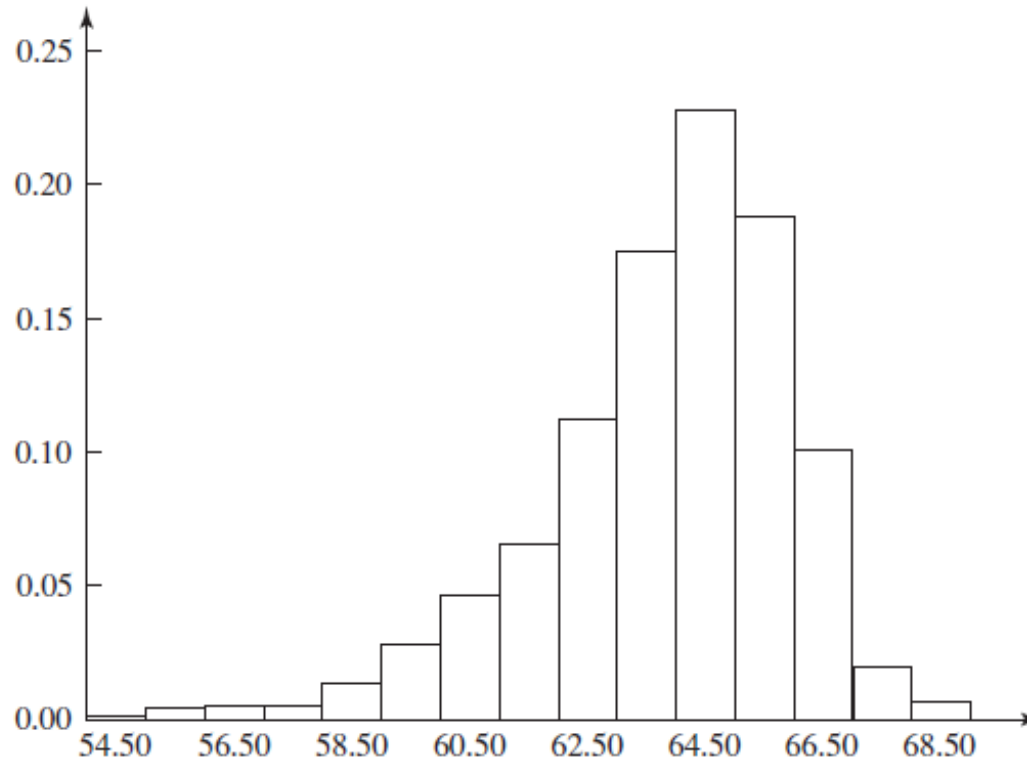


FIGURE 6.4

Histogram of the yardages for 1000 large rolls of paper for a household-products manufacturer.

Providing Input Data

- Collecting data from real system
 - Using the data directly
- Using empirical distribution
 - Generating data from the distribution
- Using theoretical distribution
 - Generating data from the distribution

Using Collected Data

- Drawbacks
 - Simulation can only reproduce what happened in the past
 - Enough data for all simulation run is often difficult
- Advantages
 - Compare a proposed system with an existing system
 - Recommended for model validation

Using Empirical Distribution

- Avoids the short-comings of using real data
 - Using good RNG's, necessary data can be generated
 - Any values between min and max can be generated
- Disadvantages
 - Might have certain irregularities
 - Especially, when data size is small
 - It is not possible to generate outside the range of observed data
 - Extreme data often determine the system performance

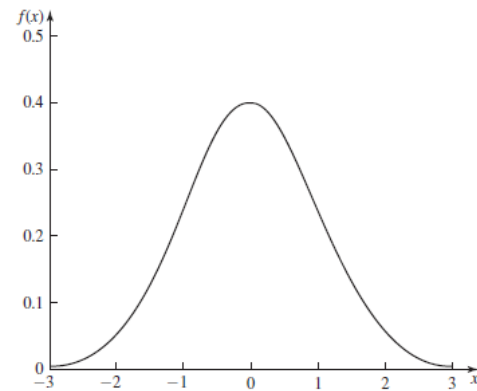
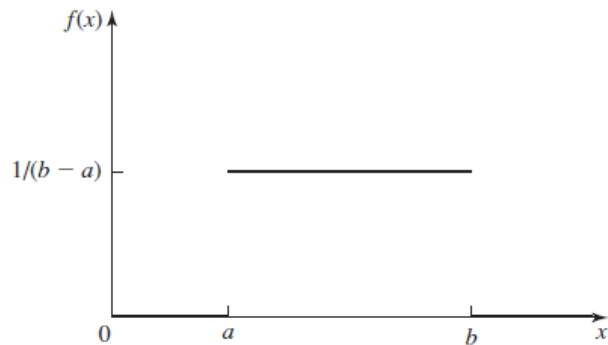
Theoretical Distribution

- Smooth out the irregularities
- Theoretical distribution can produce data outside the observed range
- Easier to change the distribution
 - e.g., for exponential the mean can be easily increased or decreased
- Theoretical distribution may not fit the input data
 - Input data are mixer of two or more heterogeneous population
 - Arbitrary large value can be generated

Useful Probability Distributions

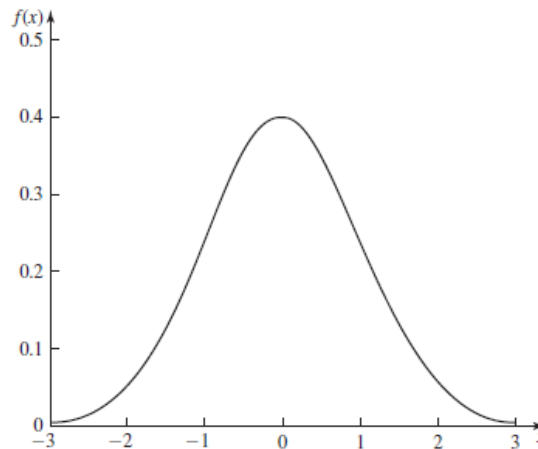
Parameterization of Continuous Distrib.

- Location Parameter (γ)
 - An x -axis location point of a distribution's range of values
 - Usually γ is the midpoint or lower end point
 - If lower end-point, known as shift parameter
 - When γ changes, the distribution shifts to left or right
 - If $\gamma = 0$, for X , and $Y = X + m$, then $\gamma = m$ for Y



Parameterization of Continuous Distrib.

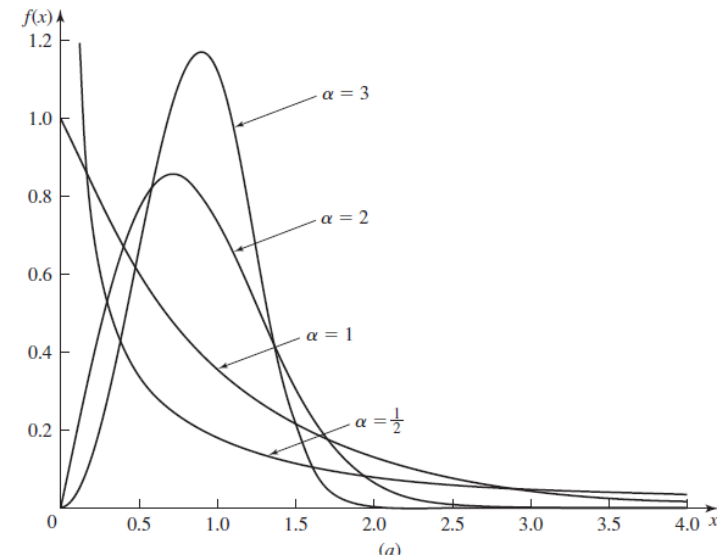
- Scale Parameter (β)
 - Scale of measurement of the values in the range of the distributions
 - A change in β compresses or expands the associated distribution
 - If $\beta = 1$, for X , and $Y = mX$, then $\beta = m$ for Y



Parameterization of Continuous Distrib.

- Shape Parameter (α)
 - Determines the basic shape of a distribution
 - A change in α generally alter a distribution properties
 - Some distribution do not have shape parameter, while other may have two.

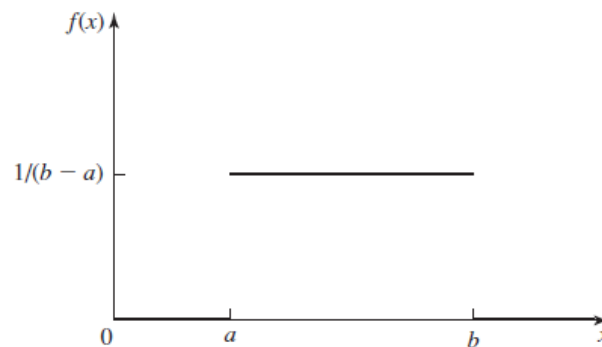
Weibull	Weibull(α, β)
Possible applications	Time to complete some task, time to failure of a piece of equipment rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.8)	$f(x) = \begin{cases} \alpha\beta^{-\alpha}x^{\alpha-1}e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$



Uniform Distribution - $U(a, b)$

Uniform	$U(a, b)$
Possible applications	Used as a “first” model for a quantity that is felt to be randomly varying between a and b but about which little else is known. The $U(0, 1)$ distribution is essential in generating random values from all other distributions (see Chaps. 7 and 8).
Density (See Fig. 6.5)	$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b < x \end{cases}$
Parameters	a and b real numbers with $a < b$; a is a location parameter, $b - a$ is a scale parameter

Uniform	$U(a, b)$
Range	$[a, b]$
Mean	$\frac{a+b}{2}$
Variance	$\frac{(b-a)^2}{12}$
Mode	Does not uniquely exist
MLE	$\hat{a} = \min_{1 \leq i \leq n} X_i, \hat{b} = \max_{1 \leq i \leq n} X_i$



Exponential Distribution - **expo** (β)

Exponential	expo(β)
Possible applications	Interarrival times of “customers” to a system that occur at a constant rate, time to failure of a piece of equipment.
Density (see Fig. 6.6)	$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Parameter	Scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	β
Variance	β^2
Mode	0

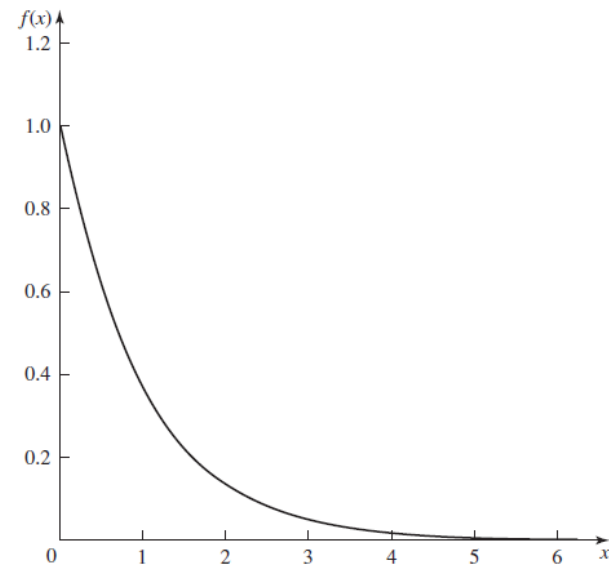


FIGURE 6.6
expo(1) density function.

Gamma Distribution – gamma (α, β)

Gamma	gamma(α, β)
Possible applications	Time to complete some task, e.g., customer service or machine repair
Density (see Fig. 6.7)	$f(x) = \begin{cases} \frac{\beta^{-\alpha} x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$ where $\Gamma(\alpha)$ is the <i>gamma function</i>, defined by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ for any real number $z > 0$. Some properties of the gamma function: $\Gamma(z + 1) = z\Gamma(z)$ for any $z > 0$, $\Gamma(k + 1) = k!$ for any nonnegative integer k, $\Gamma(k + \frac{1}{2}) = \sqrt{\pi} \cdot 1 \cdot 3 \cdot 5 \cdots (2k - 1)/2^k$ for any positive integer k, $\Gamma(1/2) = \sqrt{\pi}$
Distribution	If α is not an integer, there is no closed form. If α is a positive integer, then $F(x) = \begin{cases} 1 - e^{-x/\beta} \sum_{j=0}^{\alpha-1} \frac{(x/\beta)^j}{j!} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\alpha\beta$
Variance	$\alpha\beta^2$
Mode	$\beta(\alpha - 1)$ if $\alpha \geq 1$, 0 if $\alpha < 1$

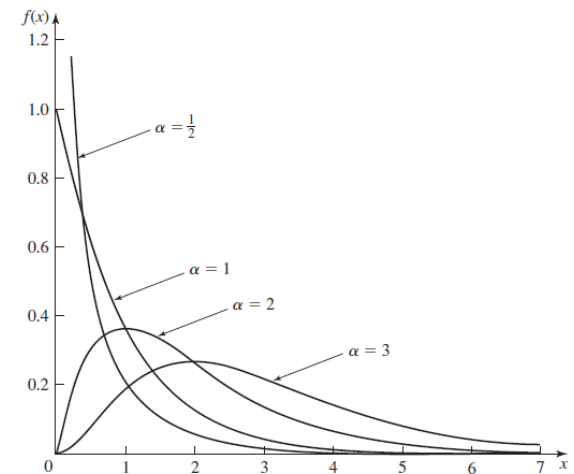
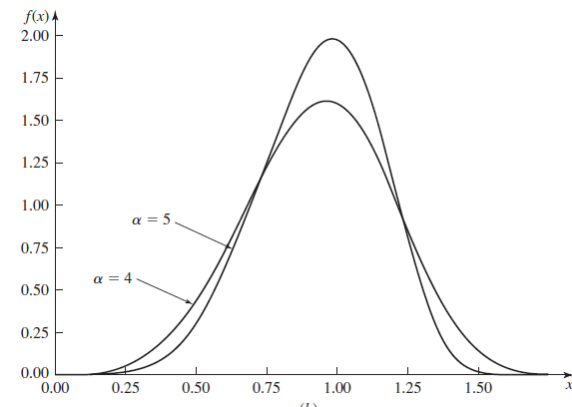
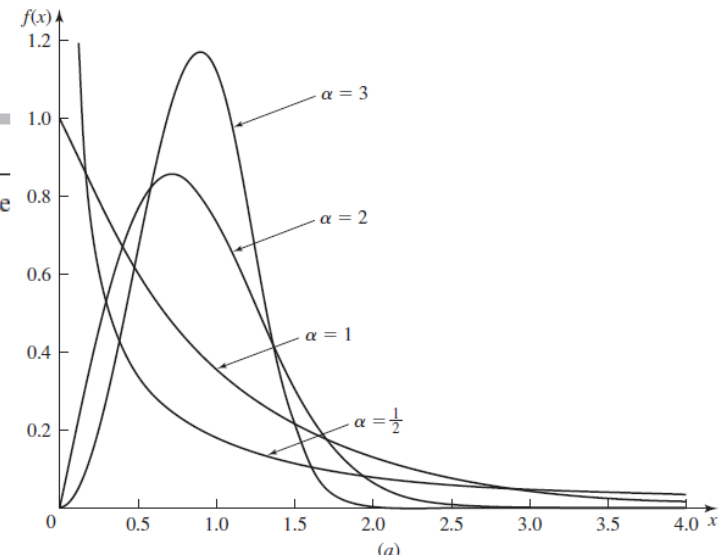


FIGURE 6.7
gamma($\alpha, 1$) density functions.

(continued)

Weibull Distribution – weibull (α, β)

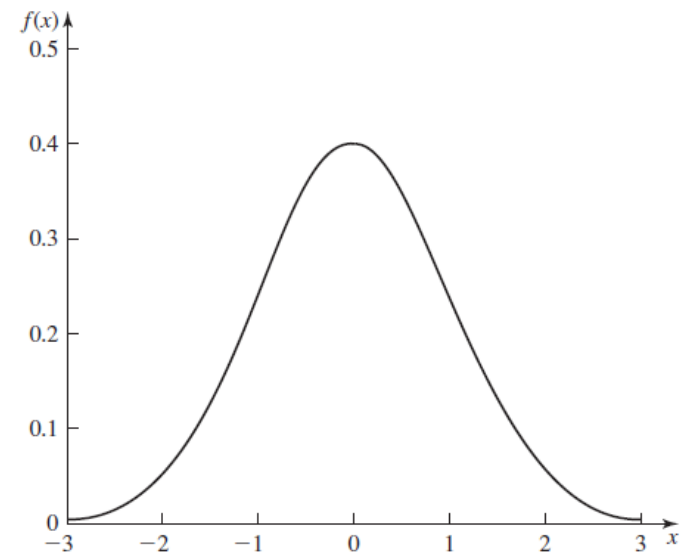
Weibull	Weibull(α, β)
Possible applications	Time to complete some task, time to failure of a piece of equipment; use rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.8)	$f(x) = \begin{cases} \alpha\beta^{-\alpha}x^{\alpha-1}e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\frac{\beta}{\alpha} \Gamma\left(\frac{1}{\alpha}\right)$
Variance	$\frac{\beta^2}{\alpha} \left\{ 2\Gamma\left(\frac{2}{\alpha}\right) - \frac{1}{\alpha} \left[\Gamma\left(\frac{1}{\alpha}\right) \right]^2 \right\}$
Mode	$\begin{cases} \beta \left(\frac{\alpha-1}{\alpha} \right)^{1/\alpha} & \text{if } \alpha \geq 1 \\ 0 & \text{if } \alpha < 1 \end{cases}$



Normal Distribution - $N(\mu, \sigma^2)$

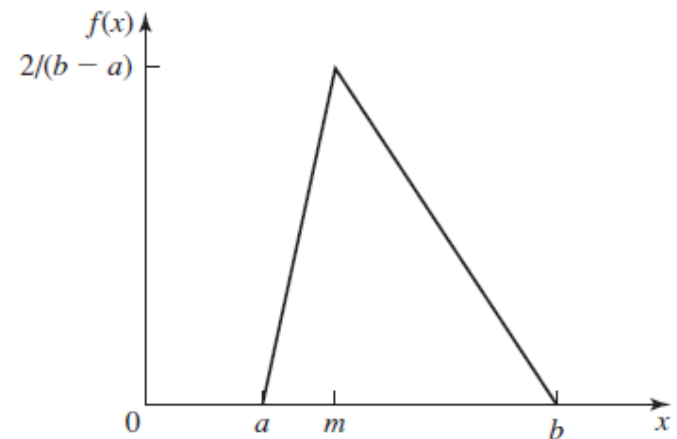
Normal	$N(\mu, \sigma^2)$
Possible applications	Errors of various types, e.g., in the impact point of a bomb; quantities that are the sum of a large number of other quantities (by virtue of central limit theorems)
Density (see Fig. 6.9)	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$ for all real numbers x
Distribution	No closed form
Parameters	Location parameter $\mu \in (-\infty, \infty)$, scale parameter $\sigma > 0$
Range	$(-\infty, \infty)$

Mean	μ
Variance	σ^2
Mode	μ
MLE	$\hat{\mu} = \bar{X}(n), \quad \hat{\sigma} = \left[\frac{n-1}{n} S^2(n) \right]^{1/2}$



Triangular Distribution

Triangular	$\text{triang}(a, b, m)$
Possible applications	Used as a rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.17)	$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(m-a)} & \text{if } a \leq x \leq m \\ \frac{2(b-x)}{(b-a)(b-m)} & \text{if } m < x \leq b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{(x-a)^2}{(b-a)(m-a)} & \text{if } a \leq x \leq m \\ 1 - \frac{(b-x)^2}{(b-a)(b-m)} & \text{if } m < x \leq b \\ 1 & \text{if } b < x \end{cases}$
Parameters	a, b , and m real numbers with $a < m < b$. a is a location parameter, b is a scale parameter, m is a shape parameter
Range	$[a, b]$
Mean	$\frac{a + b + m}{3}$



Triangular	$\text{triang}(a, b, m)$
Variance	$\frac{a^2 + b^2 + m^2 - ab - am - bm}{18}$
Mode	m

Empirical Distributions

- CDF is defined based on observed data
 - Random values are then generated from the CDF
- Depends on the availability of
 - Original data
 - $X_1, X_2, \dots, X_n$
 - Grouped data
 - Data in the form of histogram

Empirical Distribution – original data

- X_i 's are sorted into increasing order

order. Let $X_{(i)}$ denote the i th smallest of the X_j 's, so that $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$. Then F is given by

$$F(x) = \begin{cases} 0 & \text{if } x < X_{(1)} \\ \frac{i-1}{n-1} + \frac{x - X_{(i)}}{(n-1)(X_{(i+1)} - X_{(i)})} & \text{if } X_{(i)} \leq x < X_{(i+1)} \\ 1 & \text{if } X_{(n)} \leq x \end{cases} \quad \text{for } i = 1, 2, \dots, n-1$$

Figure 6.24 gives an illustration for $n = 6$. Note that $F(x)$ rises most rapidly over those ranges of x in which the X_i 's are most densely distributed, as desired.

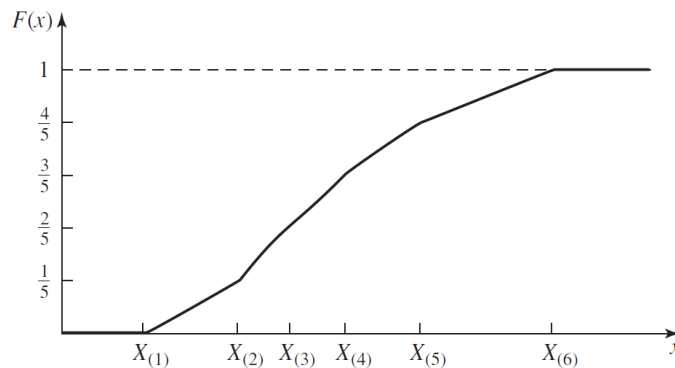


FIGURE 6.24

Continuous, piecewise-linear empirical distribution function from original data.

Empirical Distribution – grouped data

If, however, the data are grouped, then a different approach must be taken since we do not know the values of the individual X_i 's. Suppose that the n X_i 's are grouped into k adjacent intervals $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$, so that the j th interval contains n_j observations, where $n_1 + n_2 + \dots + n_k = n$. (Often the a_j 's will be equally spaced, but we need not make this assumption.) A reasonable piecewise-linear empirical distribution function G could be specified by first letting $G(a_0) = 0$ and $G(a_j) = (n_1 + n_2 + \dots + n_j)/n$ for $j = 1, 2, \dots, k$. Then, interpolating linearly between the a_j 's, we define

$$G(x) = \begin{cases} 0 & \text{if } x < a_0 \\ G(a_{j-1}) + \frac{x - a_{j-1}}{a_j - a_{j-1}} [G(a_j) - G(a_{j-1})] & \text{if } a_{j-1} \leq x < a_j \\ 1 & \text{if } a_k \leq x \end{cases} \quad \text{for } j = 1, 2, \dots, k$$

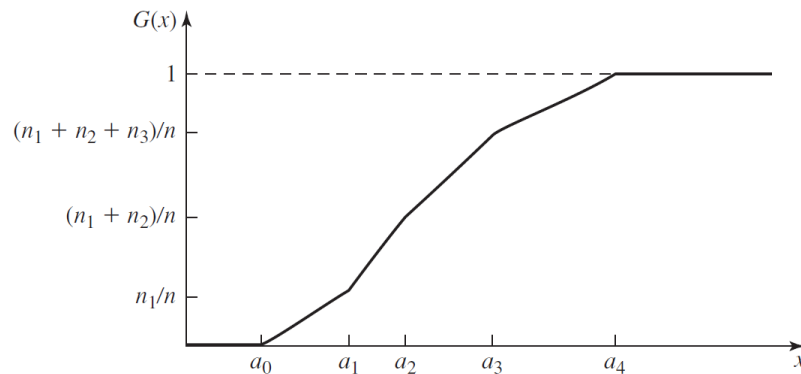


FIGURE 6.25

Continuous, piecewise-linear empirical distribution function from grouped data.